

COMPUTER SKILLS



WHAT IS A COMPUTER ?



Computer is a device can be instructed to carry out a sequences of arithmetic or logical operations automatically in seconds via a computer programming.

In general, a computer is a machine which accepts data, processes it and returns new information as output.



Data and Information



DATA : is described as some raw facts, figures and symbols, such as numbers, words, images, video and sound, given to the computer during the input phase.

INFORMATION : is a manipulate data to create information, . Information is data that is organized, meaningful, and useful. During the output Phase, the information that has been created is put into some form, such as a printed report. The information can also be put in computer storage for future use.

How does a computer work?



INPUT DATA : consists of DATA & INSTRUCTION

PROCESSING : is a set of instructions stored in the computer to carry out the instructions given by the user.

OUTPUT INFORMATION : is the set of results generated after processing the Input.



Stage of Develop the computer



- Increase the speed of the computer
- Decrease the size of computer
- Reduce the cost.
- Increase the result's accuracy.
- Ability to store huge amounts of data and information
- Facilitate to use by users.

Generations of Computers



First computer



First Generation (1940-1956) Vacuum Tubes



- ▶ Big size
- ▶ Consumes energy
- ▶ Small memory
- ▶ Used machine language
- ▶ One Operation at a time.
- ▶ Slow
- ▶ Easy to break.



Second Generation (1956-1963) Transistors



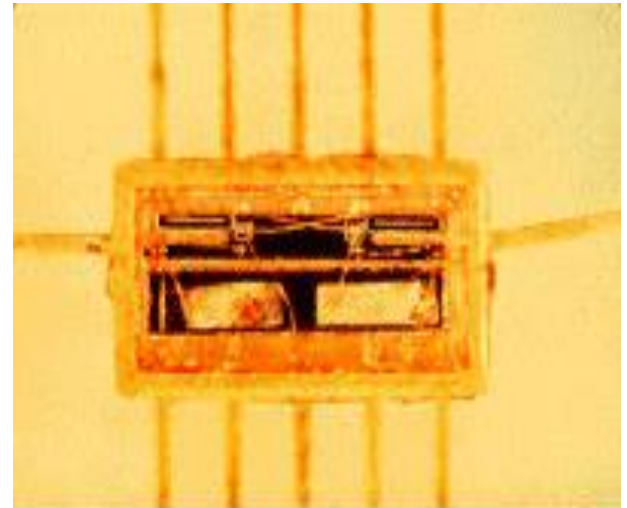
- ▶ Small size than *Vacuum Tubes*
- ▶ more energy-efficient
- ▶ faster
- ▶ Cheaper.
- ▶ Smaller.
- ▶ more reliable than their first-generation predecessors.
- ▶ Use of memory and hard drives.
- ▶ Used assembly languages



Third Generation (1964-1971) Integrated Circuits (IC)



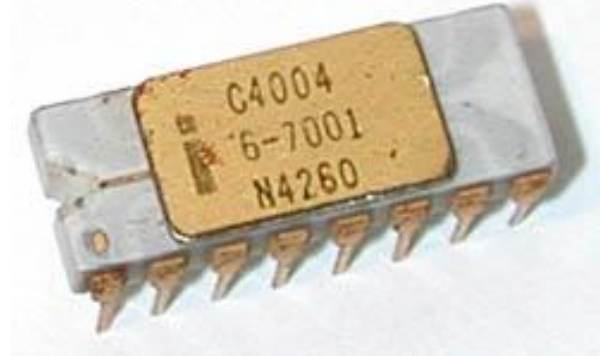
- ▶ Lightweight
- ▶ increased the speed and efficiency of computers
- ▶ Small size .
- ▶ Low cost
- ▶ Minicomputer .
- ▶ Multiprocessors
- ▶ Many tasks at a time.
- ▶ Input & output Device



Fourth Generation (1971-Present) Microprocessors



- ▶ High speed
- ▶ Pc in 1981(IBM 1981 and Apple Macintosh)
- ▶ Developed an program, Operating System and real time Systems
- ▶ Used Floppy disks
- ▶ GUI (Graphic User Interface)
- ▶ Networks & internet 1992.



Fifth Generation (Present and Beyond) Artificial Intelligence



- ▶ based on **artificial intelligence**
- ▶ Some application was developed such as **voice recognition**
- ▶ **Parallel** processing



Classifications of Computer



Divided to 3 kinds base on:

1) Work Method:

- Digital Computer.
- Analog Computer.
- Hybrid Computer.

2) Storage and accuracy in performs the tasks

1. Super computer
2. Mainframe
3. Control Computer
4. Server Computer
5. Workstation
6. Micro Computer

Classifications of Computer



3) Purpose Use :

- General purpose computers
- Special purpose computers

Classifications of Computer



Classifications of Computer

Divided to 3 kinds base on:

1) Work Method:



Analog Computer.



Digital Computer.



Hybrid Computer.

2) Storage and accuracy in performs the tasks

1



Super Computer

2



Mainframe

3



Mini Computer

4



Control Computer

5



Server Computer

6



Workstation

7



Microcomputer

3) Purpose Use :



General purpose computers



Special purpose computers

By Work Method**Digital**

Processes data in the form of digits (0 and 1).

Analog

Processes data in continuous signals.

Hybrid

Combines digital and analog processing.

By Storage & Accuracy**Supercomputer**

Largest size and highest processing power and storage capacity.

Mainframe Computer

Very large, handles massive data and supports many users.

Control Computer

Designed for monitoring and controlling industrial processes.

Server Computer

Provides services, resources or data to other computers over a network.

Workstation

High-performance computer for engineering, scientific or technical work.

Microcomputer

Smallest size, designed for individual use.

By Purpose**General Purpose**

Used to perform a wide range of common tasks.

Special Purpose

Designed for a specific task or dedicated application.

Computer Applications



Computers in the Real World

Applications across diverse sectors.



**Government
Departments**



**Banks &
Business**



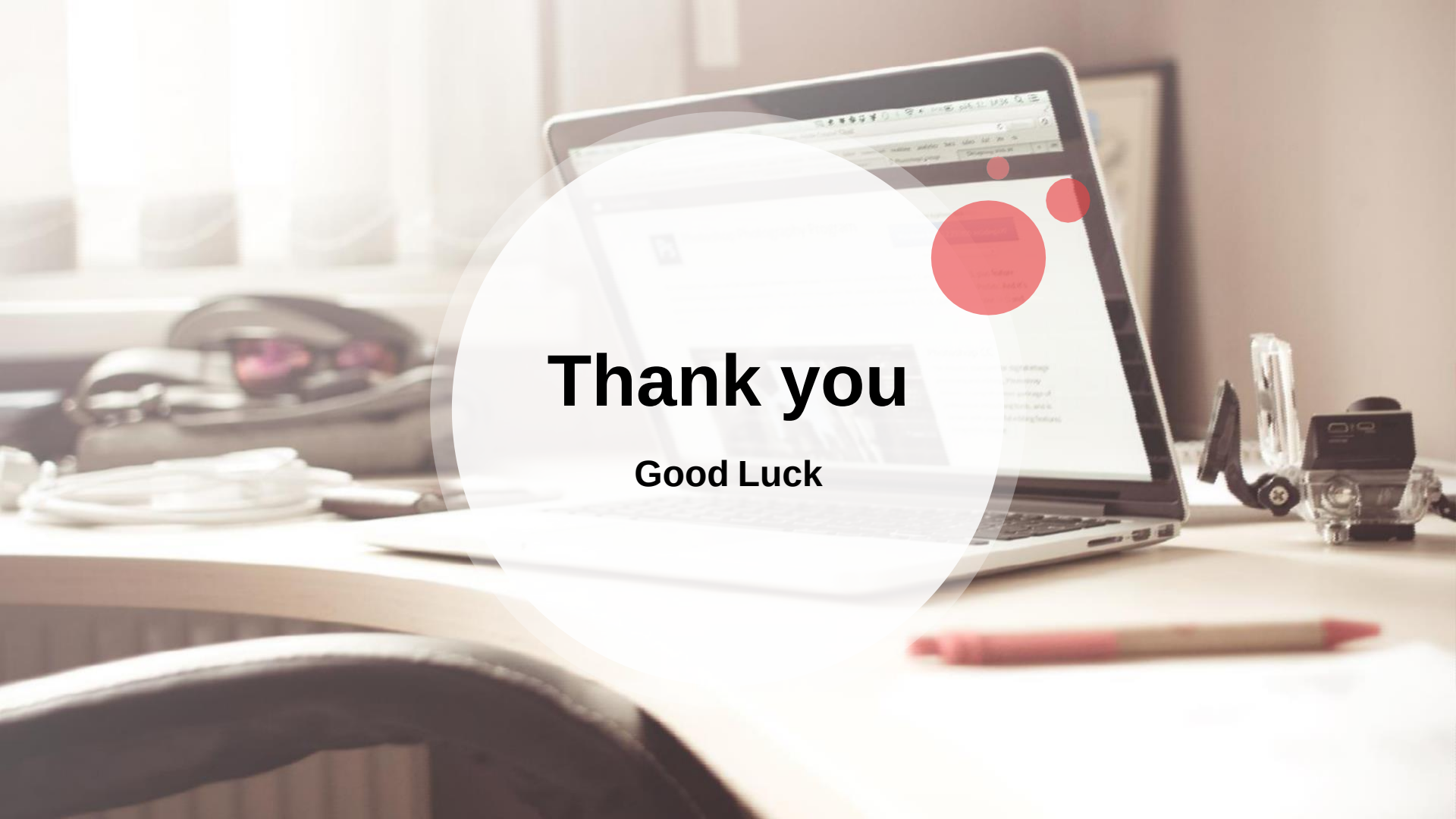
Education



**Healthcare &
Medical Uses**



Home Use



Thank you

Good Luck